Displacement Reactions

In these reactions, metals compete for oxygen or anions

- Oxidisation is the loss of electrons
- Reduction is the gain of electrons
- The more reactive metal will displace the less reactive metal from oxygen or an anion.
- If the more reactive metal has oxygen or an anion, no reaction occurs
- The bigger the difference in reactivity between the two metals, the faster the reaction

Example: $Mg + O_2 \rightarrow 2MgO$
Here magnesium is oxidised while the oxygen atom is reduced

1.5. Corrosion of Metals

Rusting: the corrosion of iron and steel to form rust (**hydrated iron (III) oxide**) via oxidation

The conditions for rusting to occur:

1. Water
2. Air containing oxygen

The Equation for Rusting of Iron

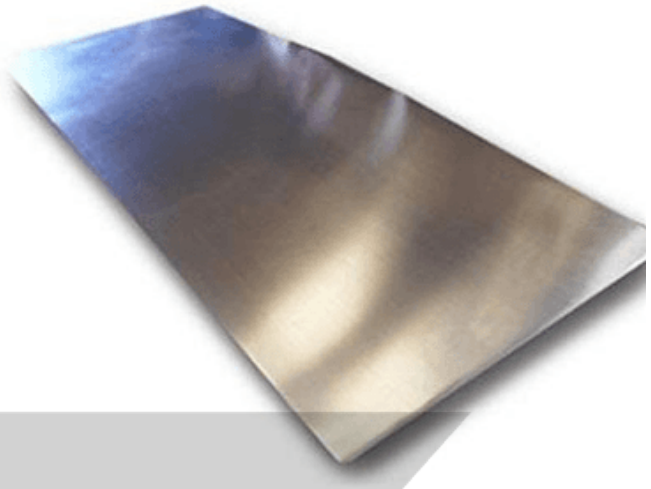
Iron + Water + Oxygen → Hydrated Iron (III) Oxide
$4Fe\ (s) + 2H_2O\ (l) + 3O_2\ (g) \rightarrow 2Fe_2O_3{\cdot}H_2O\ (s)$

Two Ways of Preventing Rust

1. **Barrier Method**
   - Coating a material to prevent the iron/steel from being in contact with water and oxygen (Painting, Greasing & Plastic Coating)
2. **Sacrificial Method**
   - When more reactive metals corrode to less reactive metals, they lose electrons in preference to iron.

Galvanisation of Iron

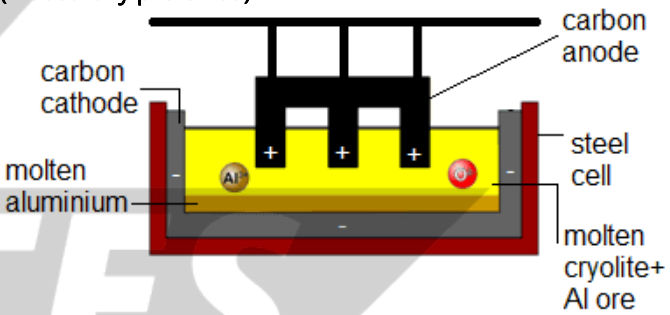
It is the **protection of iron and steel objects** by coating them with a **layer of zinc.**



The barrier method is due to the zinc layer preventing exposure to air and water and sacrificial protection due to zinc being more reactive than iron, corroding in preference to iron.

1.6. Extraction of Aluminium

- The main ore of Aluminium Oxide is bauxite
- Aluminum (III) oxide (alumina) is dissolved in molten **cryolite** (Na_3AlF_6) **– this mixture has a lower melting point (industrially preferred)**



During electrolysis, aluminium ($Al^{3+} + 3e- \rightarrow Al$) is produced at the carbon cathode, and oxygen ($2O^{2-} \rightarrow O_2 + 4e-$) at the carbon anode.

Molten Cryolite

- High temperatures are needed to melt, which is expensive. Therefore, molten cryolite **lowers the melting point/operating temperature and increases conductivity.**
- Due to the high temperature, the oxygen **reacts with the carbon** in the graphite anode to form CO_2, so the anode must **be periodically replaced.**

1.7. Extraction of Metals

- Process of separating a particular metal from its compound; metal ore

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